Modern Academy
for Engineering and Technology
Basic Science Department

Academic Year: 2021-2022
Semester: Fall
Exam Date: 30/11/2021

QUESTIONS FOR MID-TERM EXAM

Subject: (MTH 203, MTHN103, MTHn103)
Differential Equations and Transforms

Spec: Elec. and Mech.
Examiners: Assoc. Prof. Ashraf Taha & Dr. Amira Salem
Time: 60 minutes

Question 1: Find the general solution of the first ordinary differential equations

a) $y' = \frac{y}{\csc x} + \sin x.$

b) $(9xy + 10y - 2)dx + x(3x + 5)dy = 0.$

c) $\frac{x(\ln y - \ln x)}{y}\frac{dy}{dx} = (\ln y - \ln x) + 1.$

d) $(\sin x)y' + (\cos x)y = \sin x.$

Question 2: Find the general solution of the following differential equations

a) $y'''' - 5y''' + 8y''' - 6y'' = 0.$

b) $y'' + 3y' - 4y = 2e^{2x}.$

c) $y'' - 3y' + 2y = e^{2x}$, given that $y = e^{2x}$ is a homogenous solution. (**Using *Reduction of order***)

GOOD LUCK

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